

Split tenselessness: Tense, aspect, and counterfactuality in Kanien'kéha

Terrance Gatchalian^{*}
terrance.gatchalian@utoronto.ca

SULA-TripleA, UBC, 15 May 2026

1 Introduction

In discussions of the tense systems of languages with morphologically tenseless clauses, the usual narrative is an all or nothing approach: a language is taken to be *either* (covertly) tensed, or deeply tenseless

- However, as [Toosarvandani \(2025\)](#) writes: “while linguists have a habit of talking about tenseless languages, it is really clauses which do or do not have tense”

⇒ Today, I will take up this question and suggest that **Kanien'kéha**, a Northern Iroquoian language, is “tenseless” in two different senses:

- **Perfective aspect verbs** are *deeply* tenseless, making no reference to topic times
- **Non-perfective aspect verbs** are *morphologically* tenseless, making reference to topic times

To begin, Kanien'kéha event verbs appear in one of three aspectual forms.

(1) a. Perfective	b. Imperfective	c. Stative
wa'kenóhare'	ke'nóhares	wakenóhare
wa'-ke-nohare-'	ke-nohare-s	wake-nohare
FACT-1SGA-wash-PFV	1SGA-wash-IPFV	1SGP-wash.STAT
'I washed it'	'I wash it (habitually)'	'I have washed it'

^{*}**Acknowledgements.** Uncited Kanien'kéha examples are from my notes working with Konwaronhiá:wi Helen Norton, Katerí Deer, Mary Onwá:ri Tekahawáhkwen McDonald, and Akwiratékhá' Martin; *niawen'kó:wa'* to the Kanien'kehá:ka Onkwawén:na Raotitióhkwa Language and Cultural Centre team for their support and assistance. This work builds on joint work with Katya Morgunova, as well as with Willie Myers, Arihwí:saks Colin Benedict. Thank you to the many people who have commented on various aspects of this work: Luis Alonso-Ovalle, Jessica Coon, Seth Cable, Amy-Rose Deal, Michela Ippolito, Lisa Matthewson, Karin Michelson, and Lefteris Paparounas.

There are **two morphological asymmetries** that differentiate between PERFECTIVE aspect and the two non-perfective (IMPERFECTIVE and STATIVE) aspects:

- ① Non-perfective aspects are optionally followed by **tense suffixes**; perfective aspect never is (2)
- ② Perfective aspect require one of three **modal/mood prefixes** (FACTUAL, FUTURE, OR OPTATIVE), whereas non-perfective aspects require a “modalizer” suffix to co-occur with modal/mood prefixes (3)¹

(2) **Overt tense suffixes asymmetry**

- a. katá:wenskwe'
k-atawen-s-**kwe'**
1SGA-swim-IPFV-FOR.PST
'I used to swim'
- b. *wa'katá:wen'kwe'
(wa'-)k-atawen-'-**kwe'**
FACT-1SGA-swim-PFV-FOR.PST
Intended: 'I swam'

(3) **Modal/mood prefixes asymmetry**

- a. enkatá:wen'
en-k-atawen-'
FUT-1SGA-swim-PFV
'I will swim'
- b. enkatawénhseke'
en-k-atawen-hs-*(**eke'**)
FUT-1SGA-swim-IPFV-MODZ
'I will be swimming'

Previously, Baker and Travis (1997, 1998) gave an account for these morphological asymmetries by the following assumption:

(4) **Tenselessness of perfectives**

Non-perfective aspect introduces a time argument, whereas perfective aspect verbs do not.

Aim

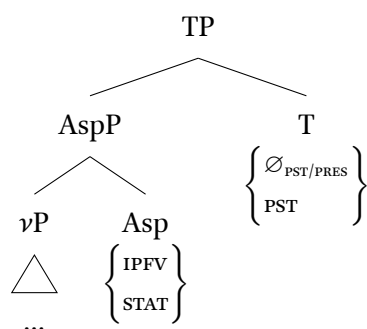
Recruit two pieces of empirical support for the hypothesis in (4):

- ① **Non-perfectives are tensed (introduce a reference time) in §2**
I will argue that non-perfective aspects are always tensed; when they do not have the optional overt tense suffix, they are *covertly tensed* (see Matthewson 2006 among many others)
⇒ This account raises the question of whether perfective forms (for which there are no overt tense suffixes) are also covertly tensed.
- ② **Perfectives are not tensed (make no reference to reference times) in §3**
I will present ongoing, joint work with Katya Morgunova, using data from **counterfactual conditionals** as a diagnostic for the lack of a tense with the perfective aspect
⇒ Thus, the absence of an overt tense morpheme means two things depending on the aspect: a *covert tense* with the IPFV/STAT aspects, and the *lack of tense* with PFV aspect.

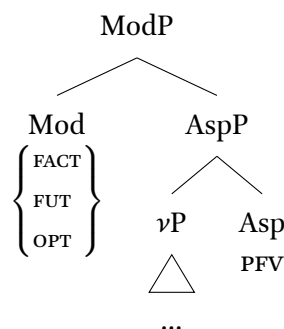
¹With non-perfective aspects, only the future and optative prefixes are possible. The modalizer is sometimes null with the perfect in Kanien'kéha but always overt in related languages.

The view we will end up with:

(5) a. Structure of non-perfective aspect verb



b. Structure of perfective aspect verb



Kanien'kéha Language Background

- Northern Iroquoian language spoken in Ontario, Québec, and New York
- by a reported 562 proficient L1 speakers and 77 advanced L2 speakers (DeCaire 2023)
- and a growing population of intermediate L2 speakers, thanks to the significant revitalization efforts including adult immersion programs (see, e.g., Stacey 2016; Gomashie 2019; DeCaire 2023).
- Uncited Kanien'kéha examples are from my notes working with Konwaronhiá:wi Helen Norton, Katerí Deer, Mary Onwá:ri Tekahawáhkwen McDonald, and Akwiratékha' Martin

2 Obligatory tense with non-perfective aspects

In this section, my aims are

- ① to argue that Kanien'kéha *imperfective* and *stative* aspect verbs have **obligatory tense**
- ② consequently, to argue that morphologically tenseless clauses involve **covert tenses**: specifically, both a covert past and a covert present

2.1 Morphological and Aspectual preliminaries

Before looking at tense, I will introduce some basic properties of temporal/aspectual meanings in the language. The basic verbal template of a verb in Kanien'kéha is given in (6).

(6) Verb template for Tense-Aspect-Modality categories

(Modal prefix) – Pronominal prefix – [verb base] – Aspect suffix (– Tense suffix)

- (7) wa'katá:wen'
wa'-k-[atawen]'-
FACT-1SGA-swim-PFV
'I swam'

(Martin 2023: 109)

Perfective aspect verbs require one of three modal prefixes.

- | | | |
|--|---|--|
| (8) a. Factual perfective
wa'kenóhare'
wa'-ke-nohare-'
FACT-1SGA-wash-PFV
'I washed it' | b. Future perfective
enzenóhare'
en-ke-nohare-'
FUT-1SGA-wash-PFV
'I will wash it' | c. Optative perfective
akenóhare'
a-ke-nohare-'
OPT-1SGA-wash-PFV
'...that I wash it' |
|--|---|--|

Table 1: Temporal interpretations of the perfective aspect with modal prefixes

Aspect	Modal	interpretation
PFV	FACT (8a)	past
	FUT (8b)	future
	OPT (8c)	modally subordinated

Non-perfective aspect verbs get the following aspectual interpretations, depending on the *telicity* of the verb stem (see Chafe 1970, 1980; Cross et al. 2023; Benedict et al. 2025; Gatchalian 2025)

- | | |
|--|---|
| (9) Interpretations of the Stative aspect | (10) Interpretations of the Imperfective aspect |
| a. watiénthon
wak-ientho-on
1SGP-plant-STAT
'I have planted.'
(telic; perfect) | a. tiénthos
k-ientho-s
1SGA-plant-IPFV
i. 'I plant.'
ii. 'I am planting.'
(habitual)
(telic; ongoing) |
| b. wakatshókwen
wak-atshokw-en
1SGP-smoke-STAT
i. 'I have smoked.'
ii. 'I am smoking.'
(perfect)
(atelic; ongoing) | b. katshókwas
k-atsohkw-as
1SGA-smoke-IPFV
'I smoke.'
(atelic; habitual) |

- Non-perfective aspects are compatible with both present and past temporal reference, regardless of their aspectual interpretation. (11) and (12) illustrate this for the ongoing interpretations of the imperfective and stative aspects.

Table 2: Aspectual interpretations of the non-perfective aspects

Aspect	telic verbs	atelic verbs
IPFV	habitual ongoing	habitual
STAT	perfect	perfect ongoing

(11) **Morphologically Tenseless Clause with Imperfective Aspect**

- a. Rató:rats.
r-atorat-s
MSGA-hunt-IPFV
'He is hunting.' (present ongoing)
- b. Oh na'kénhaton shahí:ken' **rató:rats.**
Oh na'kenhaton sh-a'-hi-ken-' ra-atorat-s
last.time COIN-FACT-1SG>MSG-see-PFV MSGA-hunt-IPFV
'When I saw him last, he was hunting.' (past ongoing)

(12) **Morphologically Tenseless Clause with Stative Aspect**

- a. Shawátis **rotshókwen.**
Shawatis ro-atshokw-en
John MSGP-smoke-STAT
'John is smoking.' (present ongoing)
- b. Shawátis shahí:ken' **rotshókwen.**
Shawatis sh-a'-hi-ken-' ro-atshokw-en
John COIN-FACT-1SG>MSG-see-PFV MSGP-smoke-STAT
'John was smoking when I saw him.' (past ongoing)

2.2 Overt tense morphology

Observation: Kanien'kéha has two overt past suffixes (see appendix for the distribution of these suffixes)

(13) **Past suffixes**

- a. rató:ratskwe'
r-atorat-s-kwe'
MSGA-hunt-IPFV-FOR.PST
'He *is/was hunting.' (Former Past -(h)kwe')

- b. rotshokwèn:ne'.
 ro-atshokw-en-hne'
 MSGP-smoke-STAT-REM.PST
 'He *is/was smoking.' (Remote Past -hne')

⇒ These past suffixes are restricted to absolute past, and cannot be used with future reference. **They are not like anterior aspects.**

(14) **No future perfects (*past-in-the-future*) with Former Past -(h)kwe'**

- a. * Nónen ó:ia' ientsóserate' tewáhsen nikahiatonhserá:ke (en)wakewennahnó:tahkwe'.
 nonen oia ientsoserate' te-wahsen ni-ka-hiatonhser-ake en-wake-wennahnot-a-hkwe'
 now next it.will.be.new.year DUPL-ten PRT-book-be.number FUT-1SGP-read-STAT-FOR.PST
 Intended: 'By this time next year, I will have read 20 books.'
- b. Nónen ó:ia' ientsóserate' tewáhsen nikahiatonhserá:ke enwakewennahnó:take'.
 nonen oia ientsoserate' te-wahsen nikahiatonhsera-ke en-wake-wennahnot-a-k-e'
 now next it.will.be.new.year DUPL-ten PRT-book-be.number FUT-1SGP-read-STAT-MODZ
 'By this time next year, I will have read 20 books.'

(15) **No future perfects (*past-in-the-future*) with Remote Past -hne'**

- a. * wísk mínit enhotsokwèn:ne'
 wísk mínit en-ro-atsokwen-hne'
 five minute FUT-MSGP-smoke-STAT-REM.PST
 Intended: 'In five minutes, he will finish smoking/have smoked.'
- b. Wísk mínit enhahsa' tsi rotshókwen
 wísk mínit en-ra-hs-a' tsi ro-atshokw-en
 five minute FUT-MSGA-finish-PFV that MSGP-smoke-STAT
 'In five minutes, he will finish smoking.'

⇒ **For the remainder of the talk, I will focus on the Former Past.**
 (See appendix for details on the Remote Past.)

2.3 Covert tenses: temporal identity under ellipsis and cessation implicatures

Overt past tense suffixes are **optional**. In collections of past oriented narratives, they are uncommon.

- More strikingly, they are not even required when the subject is marked with the "Decessive" suffix indicating that the referent is deceased; see (16)

(16) **Optionality of Past with Decessive suffix**

- a. Ake'nisten'kénha iehnén:ies(kwe').
ake-'nistenha-kenha ie-hneni-es-kwe'
FZSG>1SG-mother-DECESS F1SGA-be.tall-IPFV-FOR.PST
'My late mother was tall.'
- b. Ake'nisten'kénha athéhsa' niakononhkiwsero'tén(hne').
ake-'nistenha-kenha athehsa' ni-a-kon-onhkwi-ser-ot-en-hne'
FZSG>1SG-mother-DECESS brown PRT-FZSGA-hair-NMLZ-have.colour-STAT-REM.PST
'My late mother had brown hair.'

(17) **When the overt suffix is absent, what is the structure?**

- a. $[\text{AspP Asp}_{\{\text{IPFV/STAT}\}} [\text{vP my.mother be.tall }]]$ (tenseless)
- b. $[\text{TP T}_{\text{NONFUT}} [\text{AspP Asp}_{\{\text{IPFV/STAT}\}} [\text{vP my.mother be.tall }]]]$ (tensed with covert non-future)
- c. $[\text{TP T}_{\{\text{PAST/PRES}\}} [\text{AspP Asp}_{\{\text{IPFV/STAT}\}} [\text{vP my.mother be.tall }]]]$ (tensed with covert past/present)

(18) **Possible covert tense denotations**

- a. $\llbracket \emptyset_{\text{NONFUT}} \rrbracket = \lambda f_{\langle i, st \rangle} \lambda t' \lambda w \exists t. \boxed{\neg(t' < t)} \wedge f(t, w)$ (Matthewson 2006)
- b. $\llbracket \emptyset_{\text{PAST}} \rrbracket = \lambda f_{\langle i, st \rangle} \lambda t' \lambda w \exists t. \boxed{t < t'} \wedge f(t, w)$
- c. $\llbracket \emptyset_{\text{PRES}} \rrbracket = \lambda f_{\langle i, st \rangle} \lambda t' \lambda w \exists t. \boxed{t \circ t'} \wedge f(t, w)$

Two empirical arguments for the tensed structure in (17c)

- ① Cessation implicatures as pragmatic competition ($\Rightarrow$ covertly tensed competitor)
- ② Temporal interpretations under ellipsis ($\Rightarrow$ covert T is past or present)

2.3.1 Cessation inference

The overt past tense suffixes give rise to an inference that the eventuality does not hold at utterance time. I will call this a *cessation inference* (Musan 1997; Thomas 2012, 2014; Altshuler and Schwarzchild 2013).

(19) **Former Past gives rise to a cessation inference**

- a. Shikekhsà'a katá:wenskwe'
Shi-k-ekhsá'a ka-atawen-s-kwe'
COIN-1SGA-child 1SGA-swim-IPFV-FOR.PST
'When I was a child, I used to swim.'
 $\rightsquigarrow$ I no longer swim now.
- b. Ohna'kénhaton shahí:ken' rató:ratskwe'
Ohna'kénhaton sha'-hi-ken' r-atorat-s-kwe'
last.time COIN-1SG>MSG-see-PFV MSGA-hunt-IPFV-FOR.PST
'When I saw him last, he was hunting.'
 $\rightsquigarrow$ He is no longer hunting.

That this inference arises can be seen in (20), where it is infelicitous to use the past marked form in contexts where the eventuality is known to the speaker to continue.

(20) CONTEXT A: *Shawátis was cleaning all morning, and he's still running around cleaning as we walk into the room. (ongoing event)*

CONTEXT B: *Shawátis was cleaning all morning, but is finished now and he's sitting on the couch relaxing. (ceased event)*

- | | | | |
|----|--------------------------------------|------------------------------|---------------|
| a. | Shawátis shiiohrhòn:ke' | tehatohtáhrhos. | |
| | Shawatis shi-io-hrhonke' | te-ha-tohtahrho-s | |
| | Shawatis COIN-NP-be.morning.STAT | DUPL-MSGA-clean-IPFV | |
| | 'Shawatis was cleaning all morning.' | | (A: ✓; B: ?✓) |
| | | | |
| b. | Shawátis shiiohrhòn:ke' | tehatohtáhrhoskwe. | |
| | Shawatis shi-io-hrhonke' | te-ha-tohtahrho-s-kwe' | |
| | Shawatis COIN-NP-be.morning.STAT | DUPL-MSGA-clean-IPFV-FOR.PST | |
| | 'Shawatis was cleaning all morning.' | | (A: #; B: ✓) |

The inference with the **Former Past** is *defeasible* (21)²

(21) **Cancellation of inference via explicit denial of cessation and ignorance statement**

- | | | | | | |
|----|---|----------------------|------------------------|----------------|----------------|
| a. | Shikekhsà:a | katá:wenskwe' | tánon' | shé:kon | katá:wens. |
| | Shi-k-ekhsá'a | ka-atawen-s-kwe' | tánon' | shé:kon | k-atawen-s |
| | COIN-1SGA-child | 1SGA-swim-IPFV-PAST | and | still | 1SGA-swim-IPFV |
| | 'When I was a child, I used to swim and I still do.' | | | | |
| | | | | | |
| b. | Ohna'kénhaton | shahí:ken' | rató:ratskwe' | nek tsi | iah |
| | Ohna'kénhaton | sha'-hi-ken-' | r-atorat-s-kwe' | nek tsi | iah |
| | last.time | COIN-1SG>MSG-see-PFV | MSGA-hunt-IPFV-FOR.PST | but | NEG |
| | tewakaterièn:tare' | tóka' | shé:kon | rató:rats. | |
| | te-wak-aterientar-e' | toka' | shekon | r-atorat-s. | |
| | NEG-1SGP-know-PFV | if | still | MSGA-hunt-IPFV | |
| | 'When I saw him last, he was hunting but I don't know if he's still hunting.' | | | | |

Cessation inferences across languages have been accounted for in at least four ways, most of which rely on a tensed competitor:

- lexical account (Plungian and van der Auwera 2006)
- scalar competition account (Altshuler and Schwarzchild 2013)
- pragmatic competition account (Cable 2017)
- semantic EXH account (Kusliy and Vostrikova 2024)

⇒ I will argue here that the competition account of Altshuler and Schwarzchild (2013) extends to Kanien'kéha, given the particular profile of Kanien'kéha's tense system

²The Remote Past's cessation inference is *not defeasible* (See appendix, (46))

The basic idea:

- (22) a. [VP-IPFV-FOR.PST] $\rightsquigarrow$ past time reference
 b. [VP-IPFV- $\emptyset$] $\rightsquigarrow$ present time reference
 c. Use of FOR.PST implicates use of $\emptyset$ is not assertable
- Past-marked non-perfective aspect verbs can describe events which have ended before UT or which continue to UT (when the cessation inference is cancelled).
 - The unmarked alternative is *compatible* with present reference, and it is this *present* meaning that competes to derive the cessation inference.
 (This is how English inferences with over tenses are derived by [Altshuler and Schwarzchild 2013](#))
 - *But* unmarked alternatives in Kanien'kéha are *also* compatible with past reference.
 (See [Cable 2017](#) for related discussion on Tlingit)
 - It is crucial *precisely how* past and present reference is attained in Kanien'kéha.
 - [Altshuler and Schwarzchild](#)'s competition account requires a PRESENT tense alternative, not a NONFUTURE tense alternative
 - So far, what we've seen of Kanien'kéha is consistent with a NFUT-PAST tense system...

2.3.2 Temporal reference under ellipsis

The claim: the temporal meaning of imperfective aspect verbs is *ambiguous* between past or present, not rather than a single past-and-present (i.e., non-future).

- When the verb is elided, the temporal reference must match (a-b); this restriction is lifted if both verbs are overt (c)

(23) **Strict temporal co-reference under ellipsis (past antecedent)**

CONTEXT: *Everyone has been swimming recently to make the most of the final days of summer weather.*

Tiahià:khsera tsi náhe', Kó:r ratáwens...

Tiahià:khsera tsi náhe', Kó:r ra-atawen-s

last.week Kor MSGA-swim-IPFV

'Last week, Kor was swimming...'

- a. ... tánon' (thetén:re) Sá k ò:ni'
 tanon' thetenre Sak o'ni'
 and yesterday Sak also
 '...and (yesterday) Sak was <swimming> too.' (matching reference)
- b. #... tánon' òn:wa Sá k ò:ni'
 tanon' onwa Sak o'ni'
 and now Sak also
 '...and Sak is <swimming> now too.' (mismatching reference, elided)

- c. ... tánon' òn:wa Sák ratá:wens ò:ni'
 tanon' onwa Sak ra-atawen-s o'ni'
 and now Sak MSGA-swim-IPFV also
 '...and Sak is swimming now too.' (mismatching reference, overt)

(24) **Strict temporal co-reference under ellipsis (present antecedent)**

CONTEXT: *I come into the room and Kó:r is cleaning right now. Yesterday, I had seen Shawátis cleaning, though he is just sitting on the couch right now.*

Kó:r tehatohtáhrhos...

Kor te-ha-tohtahrho-s

Kor DUPL-MSGA-clean-IPFV

'Kó:r is cleaning...'

- a. #tánon' Shawátis ki' ò:ni'
 tanon' Shawatis ki' o'ni'
 and Shawatis just also
 '...and Shawatis <is cleaning> too' (matching reference)
- b. #tánon' Shawátis ò:ni' shahí:ken'.
 tanon Shawatis o'ni' sh-a-hi-ken-'
 and Shawatis also COIN-FACT-1SG>MSG-see-PFV
 '...and Shawatis <was cleaning> too, when I saw him.' (mismatching reference, elided)
- c. tánon' Shawátis ò:ni' shahí:ken' tehatohtáhrhos.
 tanon Shawatis o'ni' sh-a-hi-ken- te-ha-tohtahrho-s
 and Shawatis also COIN-FACT-1SG>MSG-see-PFV DUPL-MSGA-clean-IPFV
 '...and Shawatis was cleaning too, when I saw him.' (mismatching reference, overt)

A nonfuture tense incorrectly predicts that (23b) is fine.

- If Kanien'kéha morphologically tenseless clauses use a NONFUT tense, we predict that there is in fact no mismatch in the (b) cases and have no account for their infelicity.

- (25) a. [[Last.week Kó:r swim-NFUT]_α and [(yesterday) Sák <swim-NFUT> also]]
 b. $\llbracket \alpha \rrbracket^c = \lambda w \exists t \exists e [\text{Kor.swim}(e, w) \wedge t \subseteq \tau(e) \wedge \neg(t_c < t) \wedge t \subseteq \text{last.week}_c]$
 c. **There is some non-future interval over which Kor swims, and this interval is located last week.**
 d. Swapping out the adverbs, the elided clause says: **There is some non-future interval over which Sak swims, and this interval is located yesterday.**

With strict past and present tenses, the contrast in (23) is expected.

- Given an identity condition (either structural or semantic) on ellipsis, the LF in (26) is ruled out.
- The matching LF is semantically illicit, since the reference time must change for the intended meaning.

(26) * [[Last.week Kó:r swim-PAST] and [(now) Sá:k <swim-PRES> also]]

- (27) a. [[Last.week Kó:r swim-PAST]_α and [(now) Sá:k <swim-PAST> also]]
 b. $\llbracket \alpha \rrbracket = \lambda w. \exists t \exists e [\text{swim}(e, w) \wedge \text{AGENT}(e, \text{Sak}) \wedge t \subseteq \tau(e) \wedge t < t_c \wedge t \subseteq \text{last.week}_c]$
 c. **There is some past interval over which Kor swims, and this interval is located last week.**
 d. Swapping out the adverbs, the elided clause says: **There is some past interval over which Sak swims, and this interval is located now.** $\perp$

2.4 Back to cessation inferences

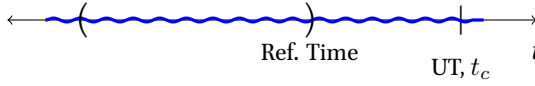
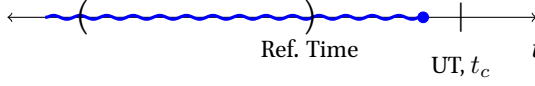
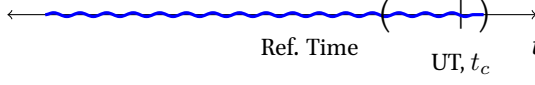
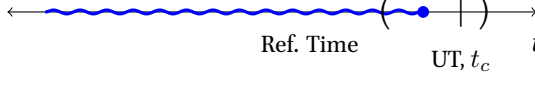
The inventory of tenses in Kanien'kéha is therefore as follows:

- (28) a. $\llbracket \emptyset_{\text{PAST}} \rrbracket = \lambda f_{\langle i, st \rangle} \lambda t' \lambda w \exists t. \boxed{t < t'} \wedge f(t, w)$
 b. $\llbracket \emptyset_{\text{PRES}} \rrbracket = \lambda f_{\langle i, st \rangle} \lambda t' \lambda w \exists t. \boxed{t \circ t'} \wedge f(t, w)$
 c. $\llbracket \text{FOR.PST}_{\text{PAST}} \rrbracket = \lambda f_{\langle i, st \rangle} \lambda t' \lambda w \exists t. \boxed{t < t'} \wedge f(t, w)$
 d. $\llbracket \text{REM.PST}_{\text{PAST}} \rrbracket = \text{t.b.d.}$

How do we derive the competition? The two competitors are imperfective aspect verbs marked with the covert present and the overt past.

- (29) a. $\llbracket \text{he.swim-IPFV-PAST} \rrbracket = \lambda w. \exists t \exists e [t < t_c \wedge t \subseteq \tau(e) \wedge \text{he.swim}(e, w)]$
 b. $\llbracket \text{he.swim-IPFV-}\emptyset_{\text{PRES}} \rrbracket = \lambda w. \exists t \exists e [t \circ t_c \wedge t \subseteq \tau(e) \wedge \text{he.swim}(e, w)]$

Note that it is not the meaning of the tenses alone, but the meaning of the tenses *and the imperfective aspect* which stand in competition. To see this, consider the following four situations:

- (30) a. Past reference time interval, continuing eventuality (Situation 1)

 b. Past reference time interval, ceased eventuality (Situation 2)

 c. Present reference time interval, continuing eventuality (Situation 3)

 d. Present reference time interval, ceased eventuality (Situation 4)


	PAST	$\emptyset_{\text{PRES}}$
Sit. 1	✓	#
Sit. 2	✓	#
Sit. 3	✓	✓
Sit. 4	#	#

Table 3: Compatibility of tenses with possible situations

Table 3 shows the felicity of the alternatives. Use of the present asymmetrically entails the past. Therefore, use of the weaker past tense imperfective form implicates that the stronger present tense imperfective form is not assertable.

2.5 Interim summary: covert tenses with non-perfective aspects

What we learned from the above diagnostics is that:

- Past-marked verbs compete with the unmarked alternative
- Unmarked verbs are covertly past or present tense (as in Blackfoot, see [Reis Silva and Matthewson 2007](#))

The resulting view is that there are two covert tenses in Kanien'kéha: the covert present competes with the overt past.

(31) Covert tense denotations (Kanien'kéha)

- a. $\llbracket \emptyset_{\text{PAST}} \rrbracket = \lambda f_{\langle i, st \rangle} \lambda t' \lambda w \exists t. \boxed{t < t'} \wedge f(t, w)$
b. $\llbracket \emptyset_{\text{PRES}} \rrbracket = \lambda f_{\langle i, st \rangle} \lambda t' \lambda w \exists t. \boxed{t \circ t'} \wedge f(t, w)$

Thus the usual past-or-present reading of morphologically tenseless clauses in Kanien'kéha is *not* due to an underspecified or weak temporal restriction (like a non-future) but from *structural ambiguity*.

$\Rightarrow$ More broadly, morphologically tenseless imperfective/stative aspect clauses are covertly tensed.

3 Tenselessness of perfective aspect: evidence from counterfactuals

Recall one of the motivating morphological asymmetries of our discussion:

(32) Overt tense suffixes asymmetry

- | | |
|--|--|
| <p>a. katá:wenskwe'
 k-atawen-s-kwe'
 1SGA-swim-IPFV-FOR.PST
 'I used to swim'</p> | <p>b. *wa'katá:wen'kwe'
 (wa'-)k-atawen-'-kwe'
 FACT-1SGA-swim-PFV-FOR.PST
 Intended: 'I swam'</p> |
|--|--|

[Baker and Travis \(1997, 1998\)](#) argued that the representation of a perfective verb lacks temporal meanings entirely, thus accounting for the lack of tense suffixes.

- If the lexicon of Kanien'kéha has covert tenses, we cannot infer from the *morphological* asymmetry in (32) that perfectives have no *syntactic* or *semantic* tense – they may simply be obligatorily covert.
- Since we know that the inventory of overt tenses depends on aspect, the perfective aspect may simply lack overt tense morphemes just as the imperfective aspect lacks the overt Remote Past suffix.

⇒ We need an independent way of determining whether perfectives are marked for tense.

3.1 Counterfactual conditionals

We will look at conditionals, and more specifically counterfactual conditionals, and argue that:

- ① Kanien'kéha uses temporal morphology to mark counterfactuals
- ② perfective aspect verbs cannot be used in counterfactual contexts
- ③ this *aspectual* restriction is a proxy for the presence/absence of *tense*, which is what determines the availability of counterfactual meanings

Brief motivating note:

- Cross-linguistically, many languages employ temporal morphology to signal counterfactuality – often restricted to past tense and/or imperfective aspect (e.g., [Iatridou 2000](#); [Bjorkman and Halpert 2017](#))
- I will argue that the restriction in Kanien'kéha is not *towards* imperfective/stative aspects, but rather *from* perfective aspects.

3.2 Central generalizations

Generalization 1: The optative prefix is necessary for counterfactual (CF) interpretations.

- Modal-less (33a) and future-marked (34a) verbs are ruled out in CF contexts

(33) **Context:** *I'm sitting in my office, tired of work.*

- a. #Sénha wakatshennón:ni' tóka' tsi kaniataratátie' tkatá:wens
 senha wak-atshennoni' toka tsi kaniatar-atatie' t-k-atawen-s
 more 1SGP-be.happy.STAT if that river-along CIS-1SGA-swim-IPFV
 Int.: 'If I were swimming by the beach, I would be happier.' (IPFV)
- b. Sénha aonkwatshennonníhake' tóka' tsi kaniataratátie' aontakatawénsheke'.
 senha aon-wak-atshennoni-hake' toka tsi kaniatar-atatie' aon-ta-k-atawen-s-heke'
 more OPT-1SGP-be.happy.STAT-MODZ if that river-along OPT-CIS-1SGA-swim-IPFV-MODZ
 'If I were swimming by the beach, I would be happier.' (OPT-IPFV)

(34) **Context:** *I am not rich, and I do not have a big house.*

- a. #tóka' enwakkwatshéheke', kanonhsowá:nen enwakenonhsó:take'
 toka en-wak-kwatshe-heke' ka-nonhs-owanen en-wake-nonhs-ot-a-ke'
 if FUT-1SGP-be.rich.STAT-MODZ 1SGA-house-be.big.STAT FUT-1SGP-house-stand-STAT-MODZ
 Int.: 'If I were rich, I would have a big house.' (FUT-STAT)
- b. tóka' aonkkwatshéheke', kanonhsowá:nen aonkenonhsó:take'
 toka a-wak-kwatshe-heke' ka-nonhs-owanen a-wake-nonhs-ot-a-ke'
 if OPT-1SGP-be.rich.STAT-MODZ 1SGA-house-be.big.STAT OPT-1SGP-house-stand-STAT-MODZ
 'If I were rich, I would have a big house.' (OPT-STAT)

Generalization 2a: The optative prefix is not sufficient for counterfactual interpretations

- Optative marked *perfective* aspect verbs cannot receive counterfactual interpretations (35a)
- Instead, the perfect (35b) or the imperfective (36) are required in counterfactual contexts, depending on the aspectual interpretation.

(35) **Context:** *There is a gathering next week, and there will be dance performances. My late mother was a very good dancer, and people loved to watch her.*

- a. #Tóka' isten'kénha tsiahià:khsera tá:we' taienónniahkwe', akwé:kon
 toka isten'-kenha tsiahiakhsera tawe' t-a-ie-nonniahkw-e' akwekon
 if my.mother-DECESS week coming DUPL-OPT-FL.A-dance-PFV all
 nón:kwe aiakaon'eskwani:'on/aiakaon'éskwen'.
 ne=onkwe a-iaka-on'esw-a-ni-'on/a-iaka-on'esw-en
 NE=people OPT-FL.O-enjoy-BEN-JR-PERF/OPT-FL.O-enjoy-BEN-PFV
 Int.: 'If my late mother was dancing next week, everyone would have liked it.' (OPT-PFV)
- b. Tóka' isten'kénha tsiahià:khsera tá:we' taiakononniáhkwen, akwé:kon
 toka isten'-kenha tsiahiakhsera tawe' t-a-iako-nonniahkw-en akwekon
 if my.mother-DECESS week coming DUPL-OPT-FL.P-dance-STAT all
 nón:kwe aiakaon'eskwani:'on.
 ne=onkwe a-iaka-on'eswa-ni-'on
 NE=people OPT-FL.O-enjoy-BEN-STAT
 'If my late mother was dancing next week, everyone would have liked it.' (OPT-STAT)

(36) **Context:** *Kor is studying to be a writer. Wari wants to marry a cook.*

- Kó:r enhahiatónseke'.
 Kor en-ha-hiaton-s-eke'
 Kor FUT-MSGA-write-IPFV-MODZ
- Toka' ahakhonniseke' ne Kór, Onwári Kó:r ahotiniá:kon
 toka a-ha-kh-onni-s-eke' ne Kor Onwari Kor a-hoti-niakon
 if OPT-MSGA-food-make-IPFV-MODZ NE Kor Onwari Kor OPT-MNSGP-marry.STAT
 'Kor will be a writer. If Kor would have been a cook instead, Wari would want to marry him!'
 (OPT-IPFV)

Generalization 2b: The optative with non-perfective aspects has free temporal reference

(37) a. *Past counterfactual*

Aonkenonhwaktani:'on tóka' thí:ken sewahió:wane' **aonkwahíá:kon.**
a-wake-nonhwakt-ani'-on toka thiken sewahiwane' **a-wak-ahi-a-k-on'**
OPT-1SGP-be.sore-BEN-STAT if this apple OPT-1SGP-fruit-JR-eat-STAT
Ioiánere' tsi iah tewakahíá:kon!
io-ianer-e' tsi iah te-wak-ahi-a-k-on
NP-be.good-STAT that NEG NEG-1SGP-fruit-JR-eat-STAT
'If I had eaten this apple, I would have gotten sick. Good thing I didn't.' (OPT-STAT)

b. *Present counterfactual*

Tóka' **aonkkwatshéheke'**, kanonhsowá:nen aonkenonhsó:take'
toka **a-wak-kwatshe-heke'** ka-nonhs-owanen a-wake-nonhs-ot-a-ke'
if OPT-1SGP-be.rich-STAT-MODZ 1SGA-house-be.big-STAT OPT-1SGP-house-stand-STAT-MODZ
'If I were rich, I would have a big house.' (OPT-STAT)

c. *Future counterfactual*

Tóka' isten'kénha tsiahià:khsera tá:we' **taiakononniáhkwen,** akwé:kon
toka isten'-kenha tsiahiakhsera tawe' **t-a-iako-nonniiahkw-en** akwekon
if my.mother-DECESS week coming DUPL-OPT-FL.P-dance-STAT all
nón:kwe aiakaon'eskwani:'on.
ne=onkwe a-iaka-on'eswa-ni-'on
NE=people OPT-FL.O-enjoy-BEN-STAT
'If my late mother was dancing next week, everyone would have liked it.' (OPT-STAT)

CF morphology

Counterfactuality is signalled by the OPTATIVE and a NON-PERFECTIVE aspect (with the modalizer):

(38) a. OPT-V-IPFV/STAT-MODZ $\rightsquigarrow$ CF

b. OPT-V-PFV $\nrightarrow$ CF

Generalization 3a: OPT+PFV may give rise to a possible future interpretation

(39) **Context:** *There is a gathering next week, and there will be dance performances. My mother is a very good dancer, and people love to watch her.*

a. Tóka' istén'ha tsiahià:khsera tá:we' **tenienónniiahkwe'**, akwé:kon nón:kwe
toka isten'ha tsiahiakhsera tawe' **t-en-ie-nonniiahkw-e'** akwekon ne=onkwe
if my.mother week coming DUPL-FUT-FL.A-dance-PFV all NE=people
eniakaon'éskwen'.
en-iaka-on'esw-en
FUT-FL.O-enjoy-BEN-PFV
'If my mother dances next week, everyone will like it.' (FUT-PFV)

- b. Tóka' istén'ha tsiahià:khsera tá:we' taienónniahkwe', akwé:kon nón:kwe
 toka isten'ha tsiahiakhsera tawe' t-a-ie-nonniahkw-e' akwekon ne=onkwe
 if my.mother week coming DUPL-OPT-FL.A-dance-PFV all NE=people
 aiakaon'éskwén.
 a-iaka-on'esw-en
 OPT-FL.O-enjoy-BEN-PFV
 'If my mother danced next week, everyone would liked it.' (OPT-PFV, ~“future less vivid”)
- c. #Tóka' istén'ha tsiahià:khsera tá:we' taiakonónniahkwen, akwé:kon nón:kwe
 toka isten'ha tsiahiakhsera tawe' t-a-ie-nonniahkw-en akwekon ne=onkwe
 if my.mother week coming DUPL-OPT-FL.P-dance-STAT all NE=people
 aiakaon'eskwanì'on.
 a-iaka-on'esw-ani'-on
 OPT-FL.P-enjoy-BEN-STAT
 Int.: 'If my mother danced next week, everyone would liked it.' (OPT-STAT)

Generalization 3a: OPT+PFV cannot get past or present interpretations

- Present interpretations are ruled out independently – perfective aspect cannot generally give rise to present readings outside of performative contexts (and with some perception verbs, yet to be tested in counterfactual contexts)
- Past readings require the perfect (as seen above), repeated in (40)

(40) **Context:** *I see a rotten apple on the counter, and throw it away.*

- a. #Aonkenonhwaktani':on tóka' thí:ken sewahió:wane' akà:iake'.
 a-wake-nonhwakt-ani'-on toka thiken sewahiwane' a-k-ahi-a-k-e'
 OPT-ISGP-be.sore-BEN-STAT if this apple OPT-ISGA-fruit-JR-eat-PFV
 Ioiánere' tsi iah tewakahíá:kon!
 io-ianer-e' tsi iah te-wak-ahi-a-k-on
 NP-be.good-STAT that NEG NEG-ISGP-fruit-JR-eat-STAT
 Int.: 'If I had eaten this apple, I would have gotten sick. Good thing I didn't.' (OPT-PFV)
- b. Aonkenonhwaktani':on tóka' thí:ken sewahió:wane' aonkwahíá:kon.
 a-wake-nonhwakt-ani'-on toka thiken sewahiwane' a-wak-ahi-a-k-on'
 OPT-ISGP-be.sore-BEN-STAT if this apple OPT-ISGP-fruit-JR-eat-STAT
 Ioiánere' tsi iah tewakahíá:kon!
 io-ianer-e' tsi iah te-wak-ahi-a-k-on
 NP-be.good-STAT that NEG NEG-ISGP-fruit-JR-eat-STAT
 'If I had eaten this apple, I would have gotten sick. Good thing I didn't.' (OPT-STAT)

3.3 Ingredients for an analysis: tense in counterfactuals

For many languages, it is (PAST) tense that does the crucial work in deriving counterfactual meanings (Iatridou 2000; Bjorkman and Halpert 2017; Mirrazi 2022; von Stechow and Iatridou 2023, among many others).

meaning	OPT-{IPFV/PERF}	OPT-PFV
past CF	✓ (37a)	#
pres CF	✓ (37b)	#
fut CF	✓ (37c)	#
possible fut	# (39c)	✓ (39b)

Table 4: Conditionals generalizations summarized

- We suggest that Kanien’kéha is part of this set of languages, though the restriction may be covert

With respect to Kanien’kéha, we would like the following ingredients for an account of counterfactual meanings:

- ① The CF marker is drawn from the temporal morphology.
 - Neither the optative nor the aspects are dedicated counterfactual markers
 - ② The CF marker is *not* the optative
 - Optative marked verbs are necessary but not sufficient for counterfactual readings
 - ③ The CF marker is *not* the aspects
 - The IPFV/STAT aspects alone are necessary but not sufficient for counterfactual readings
 - The aspects receive the same interpretations they normally receive – they are not “fake” aspects
- ⇒ The CF marker is *tense* (which correlates with the aspects)

Regardless of how we ultimately interpret the role of tense in counterfactuals (i.e., modally, e.g. Iatridou 2000; Schulz 2014; or temporally, e.g. Arregui 2007, 2009; Ippolito 2013; Mirrazi 2022), the absence of (an aspect that can co-occur with) tense means the absence of counterfactuality (see Bohnemeyer 2019 for a similar argument from Guaraní)

4 Some conclusions and consequences

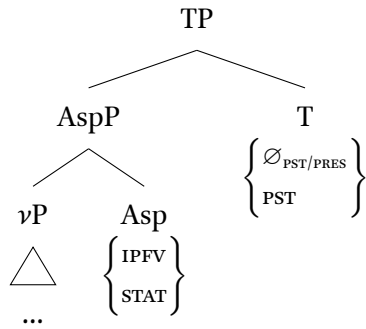
We arrived at a picture of Kanien’kéha along the following lines:

- (41) a. [ModP **Mod** [AspP Asp [_{VP} V]]] (perfective aspect)
 b. [TP T [AspP Asp [_{VP} V]]] (non-perfective aspect)
 c. [ModP **Mod** [TP T [AspP Asp [_{VP} V]]] (modalized non-perfective aspect)

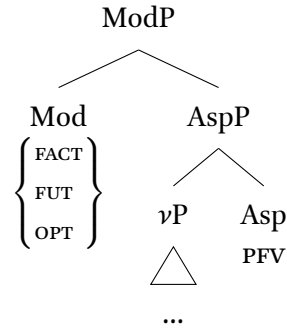
What seems to be suggested by the Kanien’kéha picture above is the close conceptual ties between the modal/mood prefixes and the tense suffixes.

- They both provide some restrictions on the index we “anchor” the eventuality to (see related ideas in [Ritter and Wiltschko 2014](#))
- Kanien’kéha seems to pick, by default, one of two ways:
 - PERFECTIVE aspect verbs: locates the eventuality at an index that precedes or follows the evaluation index (no counterfactuals)
 - NON-PFV aspect verbs: locates the eventuality at some time in the evaluation world (past or present, without modals)

(42) a. Structure of non-PFV aspect verb



b. Structure of PFV aspect verb



References

- Altshuler, Daniel, and Roger Schwarzschild. 2013. Moment of change, cessation implicatures and simultaneous readings. In *Proceedings of Sinn und Bedeutung 17*, ed. Emmanuel Chemla, Vincent Homer, and Grégoire Winterstein, 45–62. Paris: ENS.
- Arregui, Ana. 2007. When aspect matters: The case of would-conditionals. *Natural Language Semantics* 15:221–264.
- Arregui, Ana. 2009. On similarity in counterfactuals. *Linguistics and Philosophy* 32:245–278.
- Baker, Mark, and Lisa Travis. 1997. Mood as verbal definiteness in a “tenseless” language. *Natural Language Semantics* 5:213–269.
- Baker, Mark, and Lisa Travis. 1998. Events, times, and Mohawk verbal inflection. *Canadian Journal of Linguistics/Revue canadienne de linguistique* 43:149–203.
- Benedict, Arihwí:saks Colin, Terrance Gatchalian, and Willie Myers. 2025. Reversives, repetitives, and on-going telic events in Kanien’kéha. Paper presented at the 2025 Canadian Linguistics Association, June 2025.
- Bjorkman, Bronwyn, and Claire Halpert. 2017. In an imperfect world: Deriving the typology of counterfactual marking. In *Modality across syntactic categories*, ed. Ana Arregui, María Luisa Rivero, and Andrés Salanova, 157–178. Oxford: Oxford University Press.
- Bohnenmeyer, Jürgen. 2019. Counterfactuality in a tenseless language: Beyond the tense-mood confound. Presented at Tenselessness II, Universidade de Lisboa.
- Cable, Seth. 2017. The implicatures of optional past tense in Tlingit and the implications for ‘discontinuous past’. *Natural Language and Linguistic Theory* 35:635–681.
- Chafe, Wallace. 1970. *A semantically based sketch of Onondaga*. Supplement to the International Journal of American Linguistics. Baltimore: Waverly Press.
- Chafe, Wallace. 1980. Consequential verbs in the Northern Iroquoian languages and elsewhere. In *American Indian and Indoeuropean studies: Papers in honor of Madison S. Beeler*, ed. Kathryn Klar, Margaret Langdon, and Shirley Silver, 43–49. The Hague: Mouton.
- Cross, Tehokwiráthe, Terrance Gatchalian, Katya Morgunova, Willie Myers, and Ro’nikonhkátste Norton. 2023. Lexical aspect and the stative present in Kanien’kéha. In *Paper presented at the Workshop on Structure and Constituency in Languages of the Americas (WSCLA) 2023*. Montréal, QC: McGill University.

- DeCaire, Ryan Oheróhskon. 2023. The role of adult immersion in Kanien'kéha revitalization. Doctoral Dissertation, University of Hawai'i at Hilo, Hilo, HI.
- Gatchalian, Terrance. 2025. A temporal grammar of Kanien'kéha: Aspects and consequences. Doctoral Dissertation, McGill University, Montreal.
- Gomashie, Grace A. 2019. Kanien'keha/Mohawk indigenous language revitalization efforts in Canada. *McGill Journal of Education* 54:151–171.
- Iatridou, Sabine. 2000. The grammatical ingredients of counterfactuality. *Linguistic Inquiry* 31:231–270.
- Ippolito, Michela. 2013. *Subjunctive conditionals: A linguistic analysis*. Cambridge, MA: MIT Press.
- Kusliy, Petr, and Ekaterina Vostrikova. 2024. Discontinuous past: a semantic account. Paper presented at Sinn und Bedeutung 29 at the University of Göttingen.
- Martin, Akwiratékha'. 2023. *Tekawennahsonterónnion: Kanien'kéha morphology*. Kahnawake, QC: Kanien'kehá:ka Onkwawén:na Raotitióhkwa, 2nd ed. edition.
- Matthewson, Lisa. 2006. Temporal semantics in a superficially tenseless language. *Linguistics and Philosophy* 29:673–713.
- Mirrazi, Zahra. 2022. Tense in conditionals: Ins and outs. Doctoral Dissertation, University of Massachusetts Amherst, Amherst.
- Musan, Renate. 1997. Tense, predicates, and lifetime effects. *Natural Language Semantics* 5:271–301.
- Plungian, Vladimir A., and Johan van der Auwera. 2006. Towards a typology of discontinuous past marking. *Sprachtypologie und Universalienforschung (STUF)* 59:317–349.
- Reis Silva, Maria Amélia, and Lisa Matthewson. 2007. An instantaneous present in Blackfoot. In *Semantics of underrepresented languages in the americas (sula)*, volume 4, 191–214. Amherst: GLSA.
- Ritter, Elizabeth, and Martina Wiltschko. 2014. The composition of INFL: An exploration of tense, tenseless languages, and tenseless constructions. *Natural Language and Linguistic Theory* 32:1331–1389.
- Schulz, K. 2014. Fake tense in conditional sentences: a modal approach. *Natural Language Semantics* 22:117–144.
- Stacey, Kahterón:ni Iris. 2016. Ientsiwate'nikonhraié:ra'te tsi nonkwá:ti ne á:se tahaitkonhsontóntie: We will turn our minds there once again, to the faces yet to come. Master's thesis, University of Victoria, Victoria, BC.
- Thomas, Guillaume. 2012. Temporal implicatures. Doctoral Dissertation, Massachusetts Institute of Technology, Cambridge, MA.
- Thomas, Guillaume. 2014. Nominal tense and temporal implicatures: evidence from mbyá. *Natural Language Semantics* 22:357–412.
- Toosarvandani, Maziar. 2025. Languages without tense. *Language and Linguistics Compass* 19:16.
- von Stechow, Kai, and Sabine Iatridou. 2023. Prolegomena to a theory of x-marking. *Linguistics and Philosophy* 46:1467–1510.

A More on the Pasts

	form	distribution
FOR.PST	-(h)kwe'	HAB, STAT aspect
REM.PST	-hne'	STAT aspect

Table 5: Summary of distribution of past tense suffixes

- (43) a. Ieksà:a teiakonniáhkwén.
 ieksa'a te-iako-nnoniahkw-en
 child DUPL-FL.P-dance-STAT
 'The child is/was dancing'
- b. *Ieksà:a teiakonnonniahkwèn:ne'
 ieksa'a te-iako-nnoniahkw-en-hne
 child DUPL-FL.P-dance-STAT-REM.PST
 Intended: 'The child was dancing.'

- c. Ieksà:'a teiakononniahkwen**hkwe'**
 ieksa'a te-iako-nonniahkwen-hkwe'
 child DUPL-FL.P-dance-STAT-FOR.PST
 'The child was dancing.'
- b. *rowennà:note**hne'**
 ro-wennahnot-e-hne'
 MSGP-read-STAT-REM.PST
 Intended: 'He was reading.'
- (44) a. rowennà:note'
 ro-wennahnot-e'
 MSGP-read-STAT
 'He is/was reading.'
- c. rowannahnótah**hkwe'**
 ro-wennahnot-a-hkwe'
 MSGP-read-STAT-FOR.PST
 'He was reading.'
- (45) **Remote Past on Stative gives rise to a cessation inference**
- a. Shawátis **rotshokwèn:ne**.
 Shawatis ro-atshokw-en-hne
 John MSGP-smoke-STAT-REM.PAST
 'John was smoking.'
 ~> John is not currently smoking
- b. **Context:** *Paul and Sue got married in the 80s. Someone asks me whether I know Paul*
 Hen riienté:ri'. Sue **rotiniakòn:ne'**.
 Hen ri-ienteri-. Sue roti-niak-on-hne
 Yes 1SG>MSG-know-PUNC. Sue MDUP-marry-STAT-REM.PST
 'Yes, I know him. He was married to Sue.'
 ~> They are not married anymore
- (46) **Remote Past cessation inference cannot be cancelled**
- a. #Shawátis **rotshokwèn:ne** {tánon'/nek tsi} shé:kon rotshókwen.
 Shawatis ro-atshokw-en-hne tanon'/nek tsi shekon ro-atshokw-en
 John MSGP-smoke-STAT-REM.PAST and/but still MSGP-smoke-STAT
 Intended: 'John was smoking and/but he is still smoking.'
- b. #Sue **rotiniakòn:ne'** nek tsi iah tesewakaterièn:tare' tóka' shé:kon
 Sue roti-niak-on-hne nek tsi iah te-se-wak-aterien:tar-e' toka' shekon
 Sue MDUP-marry-STAT-REM.PAST but NEG NEG-REP-1SGP-know-PUNC if still
 rotiniá:ken
 roti-niak-en
 MDUP-marry-STAT
 Intended: 'He was married to Sue, but I don't know if they're still married.'